Summary Sheet

ProteinVision

Extended-Connectivity Fingerprints (ECFPs): Morgan-Based Fingerprints

Quick reference for constructing and understanding extended-connectivity fingerprints (ECFPs). These agree with the foundational methodology described by Rogers and Hahn, Extended-Connectivity Fingerprints (2010). Use our YouTube video for fuller explanations and examples.

1. Start from the molecular graph

- Represent heavy atoms as nodes and chemical bonds as edges.
- Hydrogen atoms are usually not separate nodes.

2. Assign initial atom identifiers (radius 0)

- For each heavy atom (an atom other than hydrogen), construct an initial vector containing:
  - Atomic number; atomic mass; number of heavy-atom neighbors; valence excluding hydrogens; formal charge; number of attached hydrogens; and a ring-membership flag.
- Valence excluding hydrogens is the sum of bond orders to non-hydrogen neighbors.
- Hash the vector into one integer identifier. Hashing maps the same input to the same integer. The original ECFP algorithm uses 32-bit identifiers.

3. Update identifiers using neighboring atoms

- For each update round, perform these steps for every atom:
  1. Form one pair for each neighbor: (bond-type code, neighbor's previous identifier).
  2. Sort the pairs by increasing bond-type code; break ties by increasing neighbor identifier. This removes dependence on the drawing or atom numbering.
  3. Assemble the update vector: (round number, atom's previous identifier, sorted neighbor pairs).
  4. Hash the update vector to obtain the atom's new identifier.
- Use only the previous round's identifiers for every calculation. Replace all identifiers simultaneously once the round is complete.

4. Repeat to the chosen radius

- Each update round increases the neighborhood radius by one bond. After r rounds, identifiers describe environments extending up to r bonds from each central atom.
- The number in an ECFP name is the maximum diameter in bonds (2 × radius), not the number of update rounds or the fingerprint length.

Name	Maximum radius	Update rounds
ECFP2	1 bond	1
ECFP4	2 bonds	2
ECFP6	3 bonds	3

5. Collect identifiers from all update rounds

- Retain the identifiers generated during initialization and all subsequent update rounds.
- Remove structurally redundant atom environments. Two environments may represent the same set of bonds even if their hashed identifiers differ.
- For binary fingerprints, retain each unique identifier only once.

6. Construct the binary fingerprint

- Choose the fingerprint length L (commonly 1,024 or 2,048 bits) and start with L zeros.
- For each retained identifier i (see Step 5.), calculate its bit position and set that bit to 1:
    - position = i *mod* L -> fingerprint[position] = 1
    - *mod* means the remainder after division by L. Positions run from 0 to L − 1.
- Leave all other bits at 0. Mapping identifiers into a fixed-length vector this way is called folding.

7. Important remarks and limitations

- **Presence, not frequency:** A 1 means at least one feature mapped to that position. Use a count fingerprint to record how often features occur.
- **Identifier collisions:** Different atom or update vectors can hash to the same identifier. Using 32-bit identifiers reduces this risk, but does not eliminate it.
- **Folding collisions:** Different identifiers can map to the same bit. A set bit therefore does not uniquely identify one structural feature.
- **Longer fingerprints** generally reduce folding collisions, but need more memory and storage and may increase computation. They cannot undo identifier collisions.
- **Radius vs. fingerprint length:** Radius controls how far each feature extends; fingerprint length controls how many bit positions are available.
- **Stereochemistry:** Chirality can optionally be incorporated when constructing ECFPs.

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